

Hazard/Risk Assessment

Response of a Small Mammal Population Postremediation for an Accidental Bitumen and Steam Release from a Wellhead in the Athabasca Oil Sands

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Abstract: Two releases from steam assisted gravity drainage (SAGD) wellheads occurred 3 yr apart. To track recovery of the affected areas, red-backed voles were studied 1 and 4 yr later, using population estimates, hepatic detoxification effort, body condition, and tissue metal levels as bioindicators of site recovery. From years 1 to 4, higher ethoxyresorufin-O-deethylase induction was no longer evident, capture rate was lower, and body lead residues were no longer (inversely) correlated with body condition. *Environ Toxicol Chem* 2019;38:1542–1548. © 2019 SETAC

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BACKGROUND

Among the most significant natural resources of northern Alberta, Canada, is the extensive oil sands deposits. Because of the nature of these deposits, including their depth, it is estimated that only 20% can be extracted using open-pit mining methods (Canadian Association of Petroleum Producers 2018). The remaining deeper reserves are only recoverable by in situ extraction methods which use drilling, steam injection, combustion, or other sources of heat injected into the reservoir to warm the bitumen (oil-impregnated sand) so that it can be pumped to the surface through recovery wells. In July 2010 and once again in July 2013, erosional wear at a producing well head resulted in the release of process-affected water, steam, and bitumen. The 2010 incident released 259 m³ of bitumen and 761 m³ of water. Of this, approximately 75% was collected on the well pad, whereas the remaining 25% (65 m³) was deposited on the surrounding vegetation, encompassing 636 ha. A second, smaller release occurred in 2013 when estimated fluid and steam volumes released were 58.2 m³ of material that affected approximately 63 ha of land. Deposition of bitumen and process-affected water on vegetation was estimated to vary in thickness from <0.1 to 0.7 mm. There is little in the published literature that describes similar wellhead releases, by way of comparison. Berger et al. (2016) provide a detailed description

of the deposition of crude bitumen and water resulting from these spills, with varying low-impact zones defined based on the amount of bitumen covering the understory vegetation. Bitumen coating for the 2010 release ranged from >75% in the high-impact zone to 25 to 75% in the moderate-impact zone to <25% in the low-impact zone. Bitumen coating for the 2013 release ranged from 20 to 50% in the high-impact zone to <10% in the moderate-impact zone to <3% in the low-impact zone.

Remediation at a spill site is unique to each situation. The cleanup method recommended to the Alberta Energy Regulator by Devon Canada Corporation was to selectively clear all of the vegetation and terrestrial lichen preferred by herbivorous wildlife, concentrating on the areas that had been more heavily bitumen-coated, which would serve to minimize future impacts to the site. This alternative strategy for cleanup entailed removing the most heavily coated vegetation from the 2010 release pad (120 × 10 m) and hand-clearing deciduous vegetation and terrestrial lichens in the 2010 high-impact zone. This served to remove potential forage for ungulates and snowshoe hare (*Lepus americanus*). In addition, absorptive materials were used to remove surface slicks on and around bodies of water. The standard approach would be to remove all vegetation and heavily impacted soil, effectively removing all wildlife habitat. The alternative approach resulted in much of the area being left intact for wildlife, though accepting that a certain level of remaining contamination may occur in an otherwise intact ecosystem. This was anticipated to be less harmful and permit a more rapid recovery of the site/ecosystem compared with complete removal of all vegetation.

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After the cleanup, ecosystem monitoring included wildlife surveys (ungulate pellet/browse surveys and winter track counts) and an ecological risk assessment (ERA), which was completed in 2012 and 2013. The purpose of the ERA was to evaluate the potential for toxicologically induced changes in health that ecological receptors (flora/fauna) living in or passing through the affected area may experience. The ERA was conducted according to a widely recognized framework that progressed from a qualitative initial phase (problem formulation) through exposure and toxicity assessments and culminating in a quantitative/qualitative risk characterization and discussion of the assumptions and uncertainties inherent to ERA. To support and elaborate on the conclusions made for the ERA, a small mammal trapping program was initiated in 2014, with southern red-backed voles (*Myodes gapperi*) as the target species. Southern red-backed voles were chosen because they are indigenous to the area, were likely to be exposed to contaminants of potential concern (COPC) because of their home range size of <0.5 ha (Merritt and Merritt 1978) and herbivorous diet, and provide a practical and efficient monitoring candidate and because selective vegetation removal did not target southern red-backed forage. The present study demonstrated that voles in different impact zones across the release sites were equally present and had no observable differences in morphology or in COPC blood levels that could be directly related to bitumen exposure. Although a significant increase in cytochrome P450 (CYP1A1; hepatic detoxification) activity was observed in specimens collected in the high-impact zones when compared to reference sites, which is indicative of exposure to contaminants, it was not correlated with any observable adverse biological outcome (Berger et al. 2016). In 2017, a 3-yr follow-up capture program was carried out, targeting the same species to determine if recovery (i.e., measurable positive or negative changes) was still ongoing or if the populations had re-established background or reference levels. Specifically, the objectives of the present study were to compare the findings in resident voles from 2014 and 2017 as they pertained to capture rates (i.e., relative abundance), ethoxyresorufin-O-deethylase (EROD), tissue metals, and morphological measurements.

MATERIALS AND METHODS

Field collection of animals and morphometric data

After the accidental bitumen releases in 2010 and 2013, 2 yr of sampling (2014 and 2017) of southern red-backed voles were carried out for the present study. Five specimens were targeted for each of 6 bitumen-affected regions: 2 high-impact, 2 medium-impact, and 2 low-impact areas plus 2 reference areas. The 3 impact zones were delineated based on the percentage of bitumen-covered understory vegetation, through helicopter reconnaissance, aerial photographs, and ground surveys. The 2010 high-impact zone had >75% coverage, the medium-impact zone had 25 to 50% coverage, and the low-impact zone had <25% coverage (Amec Earth and Environmental, Calgary, AB, Canada, unpublished data); and the 2013 high-impact zone had >20 to 50% coverage, the medium-impact zone had <10% coverage, and the low-impact zone had <3% coverage (HAB-TECH

Environmental, Calgary, AB, Canada, unpublished data). The reference sampling areas were selected outside of the impact zones in similar habitat types (Supplemental Data, Figure S1).

The sampling sites in 2017 were used to replicate areas previously sampled in 2014 (detailed in Berger et al. 2016). Sherman live traps baited with a mixture of peanut butter and rolled oats placed on a cracker were used to capture the specimens in impacted and reference sites (Supplemental Data, Figure S2). Traps were outfitted with polyester bedding and encased in a 2-L milk carton to provide thermal and moisture protection for live specimens. Traps were placed in a linear transect/grid spaced approximately 10 m apart, replicating previous sample sites and adding sites in areas of low capture rates or trap destruction by interfering wildlife. Traps were checked twice daily.

Captured voles were anesthetized using isoflurane and then euthanized via cervical dislocation (Garron et al. 2012). All trapping/handling and subsequent euthanasia processes followed Canadian Council on Animal Care guidelines (2010) and were approved by Alberta Environment and Parks and carried out under the authority of research permit 57820 and collection license 57822. Morphological measurements of the captured animals were recorded, including body length, body mass, tail length, hindfoot length, and ear length. Overall body condition was subjectively assessed by the field biologist using professional judgment based on the condition of the pelt (skin lesions, bare patches, cysts, wounds), musculoskeletal structure (broken bones, lameness, deformities), healthy mass (no prominent ribs), activity level, and any other potential signs of illness (crusting around eyes or nostrils, brightness of eyes); and sex was determined. The total number of trap nights and capture rate were recorded. The dissection of specimens was completed in 2 stages. First, liver samples were collected, weighed, flash-frozen, and stored in liquid nitrogen within 10 min of death for EROD analysis. Second, a minimum of 5 g of organ and muscle tissue was collected for the analysis of metals. These samples were stored on ice and subsequently frozen between –18 and –20 °C.

Population density estimates

Trapping was completed in early spring after snowmelt, with the aim of capturing adult specimens only, to provide a consistent relative population estimate and to capture animals with the greatest potential exposure to contaminants. The 2014 sampling occurred from 14 to 20 May and that in 2017, from 12 to 26 May. Relative population estimates were derived from captures per 100 trap nights (total number of captures divided by trapping effort). A trap night is 1 trap set for 24 h.

Scaled mass index to determine condition factor

Body condition of the red-backed voles was calculated using the scaled mass index (SMI). Based on studies that include mammals, birds, and reptiles, the SMI is a more representative indicator of the relative energy reserves and other body components than the more widely and traditionally used regressions of body length versus mass (Peig and Green 2009). The SMI of body condition (Mi) provides a more accurate

prediction of an animal's health and fitness and best explains that fraction of mass associated with structural size. In this population, body length without tail was used as a proxy of skeletal size. The SMI was calculated as follows: $SMI_i = M_i(L_0/L_i)^{bSMA}$. Scaled mass index, body mass, and tarsal length of the individual (i), are SMI_i , M_i , and L_i respectively; $bSMA$ is the slope over Pearson's r . The reference length, L_0 , was the mean tarsal length of all voles measured in 2014 to 2017.

Hepatic EROD analysis

The liver serves as a major organ for biotransformation of pollutants and other foreign compounds such as those produced by plants to protect themselves against herbivory, to facilitate their excretion as efficiently as possible. This process is a protective mechanism meant to detoxify compounds, which is followed by enhanced excretion. Activity of the hepatic enzyme EROD is commonly used to determine the detoxification efforts by organisms, which is assumed to reflect the level of contaminant challenge being dealt with by the animal. The EROD assays for the voles in the present study (performed by the Smits lab) were carried out using the methods detailed in Berger et al. (2016; which were also carried out by the Smits lab). All tissue handling and analyses were conducted on ice, with chilled equipment and reagents. Briefly, a 0.5- to 2-g piece of frozen liver was weighed, semithawed, minced, and transferred to a pre-chilled tissue homogenizer for homogenization and preparation of postmitochondrial supernatant for microsome fractionation. The microsomal pellet was resuspended with resuspension buffer, and a 30- μ L aliquot was removed for determination of the protein concentration using the Pierce BCA Protein Assay Kit (product 23 227). A pooled sample of the microsomal suspension was used to establish conditions for EROD determination; 20 μ g of protein per well, 6 μ M 7-ethoxyresorufin (final in reaction), and 2 mM nicotinamide adenine dinucleotide phosphate (NADPH; final concentration) with an incubation time of 17 min at 37 °C was required. Twenty μ L (~20 μ g) of microsomes diluted with 113 μ L of the 7-ethoxyresorufin solution were added to the plate, which was incubated in the dark at room temperature for 2 min; then 67 μ L of NADPH (2 mM) was quickly added to all of the wells. Plates were subsequently incubated at 37 °C in the plate reader for 17 min. The reaction was stopped by adding 30 μ L acetonitrile. The reaction was read on a fluorescent plate reader at 530 nm excitation and 590 nm emission. The same diluted samples were used to measure protein content with the Pierce BCA kit.

Metal burden analysis

A standard suite of 30 metals was analyzed with inductively coupled plasma mass spectroscopy by Maxxam Analytics on vole tissue samples from both 2014 and 2017. Standard quality-assurance and -control procedures ensured reliable results (vole tissue metal analysis is shown in Supplemental Data, Tables S1 and S2). A total of 5 g wet weight was required for each sample. To achieve the laboratory request for 5 g per sample and to prevent bias because of using different tissues from the voles,

both kidneys plus the remaining liver, heart, lung, and muscle tissue were collected for each specimen. The samples were digested in nitric acid/hydrogen peroxide to remove the organic matrix and aid in dissolution of the metals, which were maintained in solution through a final hydrochloric acid step.

Statistical analyses were carried out using IBM SPSS statistics 23. Data were analyzed using univariate general linear models. Normality and homogeneity of variances of the data were tested through the Shapiro-Wilk test and Levene's test, respectively. Non-normal data were log- or ln-transformed as required. For the outcomes where sex was determined to be a significant covariate, males and females were analyzed separately. "Year" was identified as the fixed factor for comparison of body condition or residue levels in the voles between years. Depending on the normality of the data distribution and group size differences, one-way analysis of variance with Tukey's tests or least significant difference post hoc tests were used to distinguish significant differences among the exposure groups. Significance was set at $p < 0.05$.

RESULTS

All presented methods and results are for the present study exclusively, with reference to Berger et al. 2016 only as a comparison or to provide background information or comparison.

During the 2014 survey, a total of 51 southern red-backed voles and 1 deer mouse (*Peromyscus maniculatus*) were captured. Forty southern red-backed voles were analyzed for metal body burden and EROD; an additional 11 captures were released alive. The overall capture rate per 100 trap nights was very similar in the impacted and reference areas (Table 1).

In the follow-up survey in 2017, a total of 55 southern red-backed voles, 9 deer mice, and 3 shrew species (*Sorex* spp.) were captured. Again, 40 voles were tested for body burden of metals, EROD, and body condition (SMI), although an additional 7 red-backed voles were analyzed exclusively for metal residues, with 8 released alive. The overall capture rate was markedly lower in 2017 than in 2014. In 2017, the capture rate was somewhat higher on the impacted sites than in the reference areas (Table 1).

Morphological characteristics were recorded for all voles, which were then analyzed for EROD and/or metal body burden. Body mass, body length, tail length, hindfoot length, and ear length were consistent across the 3 impact zones and reference areas. Body mass and length were slightly higher in 2014 than 2017 but within the normal range of variability for the species (Banfield 1974; Tables 2 and 3). All specimens were considered to be in "good" body condition as judged by the field biologist.

Some difficulty arose in detecting patterns or differences among exposure zones when using 4 categories of exposure (3 with different intensities of oil-related contamination plus 1 reference site, each year). This was in part because of low numbers; for example, zones 2 (low) and 3 (moderate) had only 3 female voles per site in 2014, with none in zone 2 in 2017. This, together with the finding that EROD activity was not

TABLE 1: Southern red-backed vole capture rates at impacted and reference sites

Capture site	2017 Captures per 100 trap nights	2014 Captures per 100 trap nights
2010 high impact	5.8	12.5
2013 high impact	6.5	15.8
Total high impact	6.1	15.4
2010 medium impact	2.3	5.6
2013 medium impact	5.1	25.8
Total medium impact	3.0	10.7
2010 low impact	11.7	10.0
2013 low impact	2.7	8.1
Total low impact	4.6	8.8
Overall impacted	4.1	10.8
Overall reference	3.4	10.6

different between zones 2 and 3 for either males or females, made it practical and defensible to combine exposure levels 2 and 3, creating 1 “moderate”-exposure zone for the balance of the analyses.

Comparing the EROD responses between years for males and females together, only the highly impacted area 3 was significantly different from reference area in 2014 ($p < 0.015$), a difference that was no longer evident in 2017 (Figure 1).

In 2014, based on reference (1), plus moderate- (2), and high- (3) exposure zones, females had significantly higher EROD activity than males at the reference sites ($p = 0.004$) as well as at the low- and medium-exposure zones ($df = 1$, $F = 18.534$, $p = 0.0001$) but not at the high-exposure sites ($p = 0.35$). Because of the sex difference in EROD response in 2014, EROD was analyzed separately for males and females. Females in the highest exposure zone had higher EROD activity than those from the reference site ($p = 0.005$) and those from the low- to moderate-exposure sites ($p = 0.004$; Fisher's least significant difference post hoc test).

Similarly, for the males, voles from the highest-exposure sites had higher EROD than those from the reference and moderate-exposure zones (both $p = 0.007$).

In 2017 there were no longer differences in EROD activity among the exposure groups or compared with the reference site, for the females ($p = 0.56$) or males ($p = 0.30$).

Regarding body condition (SMI), whether males and females were considered together or separately, in each year, the SMI was not different among the 3 exposure sites (reference, moderate, high). However, from 2014 to 2017, differences in body condition emerged. In 2014, there was no difference in the SMI of the female ($p = 0.61$) or male ($p = 0.87$) voles between the reference site and the different exposure zones (Figure 2A). However, comparing the population with site as a covariate, the SMI was significantly lower in 2017 than in 2014 ($p = 0.017$; Figure 2B).

Considering each year separately, a relatively small subset of the metals and metalloids were higher in the voles from the oil-exposed zones compared with those from the reference sites, differences which changed over the 3-yr gap by the second sampling. No metals and metalloids were at or near toxic levels. Lead (Pb) residues in the voles were different among sites, a difference that was evident in both 2014 and 2017. Considering all of the animals, tissue Pb was higher in voles from both the moderate- ($p = 0.015$) and high- ($p = 0.006$) impact zones relative to the reference site in both capture years (Figure 3).

Examining the years separately, in 2014 Pb levels in the moderately and highly exposed zones were not different from each other, but Pb was higher than in the reference-zone animals ($p = 0.025$), an increase which persisted in 2017 ($p = 0.038$).

Lead residues appeared to negatively affect body condition, a correlation that differed between years. In 2014, Pb was negatively correlated with body condition considering all animals (Pearson correlation 2-tailed $p = 0.04$), a correlation which was stronger when only looking at oiled sites ($p = 0.015$; Figure 4A and B). By 2017, Pb was no longer correlated with SMI (all sites $p = 0.26$, oiled sites $p = 0.65$).

For the other 29 metals measured, only antimony (Sb) and chromium (Cr) were higher than in the reference-site voles in the most heavily exposed zone. In 2014 Sb was marginally higher ($p = 0.06$), whereas Cr was significantly higher ($p < 0.03$) than in the reference-site animals and SMI was positively

TABLE 2: Southern red-backed voles' morphological measurements 2014

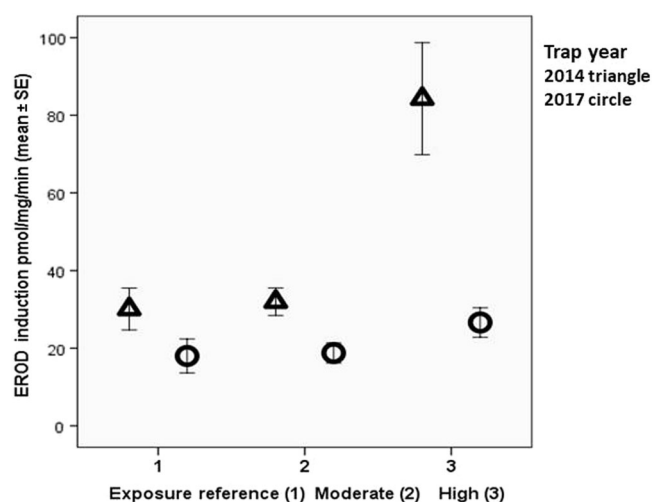
Capture site	Body mass (g)			Body length (mm)			Tail length (mm)			Hindfoot length (mm)			Ear length (mm)		
	<i>n</i>	Mean	SE	<i>n</i>	Mean	SE	<i>n</i>	Mean	SE	<i>n</i>	Mean	SE	<i>n</i>	Mean	SE
2010 high impact	5	19.3	0.8	5	90.7	3.8	5	30.8	0.6	5	16.6	0.6	5	13.9	0.6
2013 high impact	5	18.9	1.3	4	92	3.5	5	35.7	3.3	5	17.4	0.5	5	15.5	0.4
Total high impact	5	19.1	0.7	9	91.4	2.5	5	33.3	1.8	5	17	0.4	5	14.7	0.4
2010 medium impact	5	23.7	0.8	5	98	2.2	5	34.5	1.6	5	17.4	0.4	5	16.3	0.3
2013 medium impact	5	17.3	0.5	5	94.8	1.0	5	32.8	0.5	5	16.4	0.4	5	15.8	0.5
Total medium impact	5	20.5	1.2	5	96.4	1.3	5	33.7	0.8	5	16.9	0.3	5	16.1	0.3
2010 low impact	5	20.9	1.0	5	94.5	2.2	5	34.2	0.9	5	17.5	0.7	5	15.5	0.8
2013 low impact	5	20.8	1.3	5	96.8	1.4	5	31.3	0.6	5	16.1	0.5	5	16	0.6
Total low impact	5	20.9	0.8	5	95.7	1.3	5	32.8	0.7	5	16.8	0.5	5	15.8	0.5
Overall impacted	30	20.2	0.5	29	94.5	1.0	30	33.2	0.7	30	16.9	0.2	30	15.5	0.2
Overall reference	10	20.2	1.2	10	97	1.6	9	33.9	2.8	10	16.5	0.3	10	14.9	0.3

Source: Berger et al. (2016).
SE = standard error.

TABLE 3: Southern red-backed voles' morphological measurements 2017

Capture site	Body mass (g)			Body length (mm)			Tail length (mm)			Hindfoot length (mm)			Ear length (mm)		
	n	Mean	SE	n	Mean	SE	n	Mean	SE	n	Mean	SE	n	Mean	SE
Pad B high impact	5	18.2	1.1	7	90.1	3.6	7	32.7	1.1	7	16.6	0.4	7	14.9	0.4
Pad C high impact	5	17.5	1.3	6	90.6	2.8	6	33.4	0.6	6	17	0.6	6	14.6	0.5
Total high impact	10	17.9	0.8	13	90.4	2.2	13	33.1	0.6	13	16.8	0.3	13	14.8	0.3
Pad B medium impact	5	18.7	1.1	7	92.3	1.9	7	33.2	0.6	7	16.4	0.4	7	14.7	0.4
Pad C medium impact	5	17.3	0.8	6	91.1	1.5	6	32.5	1.1	6	16.5	0.4	6	14	0.4
Total medium impact	10	18	0.7	13	91.7	1.2	13	32.9	0.6	13	16.5	0.3	13	14.4	0.3
Pad B low impact	5	17.3	0.9	6	89.1	2.9	6	32.2	1.6	6	15.7	0.6	6	14.5	0.5
Pad C low impact	5	18.1	0.6	5	95.2	2.3	4	32.8	0.8	5	17.1	0.2	5	14.8	0.2
Total low impact	10	17.7	0.5	11	92.2	2.0	11	32.5	1.0	11	16.4	0.4	11	14.7	0.3
Overall impacted	30	17.9	0.4	37	91.4	1.1	36	32.8	0.4	37	16.6	0.2	37	14.6	0.2
Overall reference	10	18.6	0.9	10	90.9	1.2	10	34.4	0.4	10	16.7	0.2	10	14.7	0.3

SE = standard error.

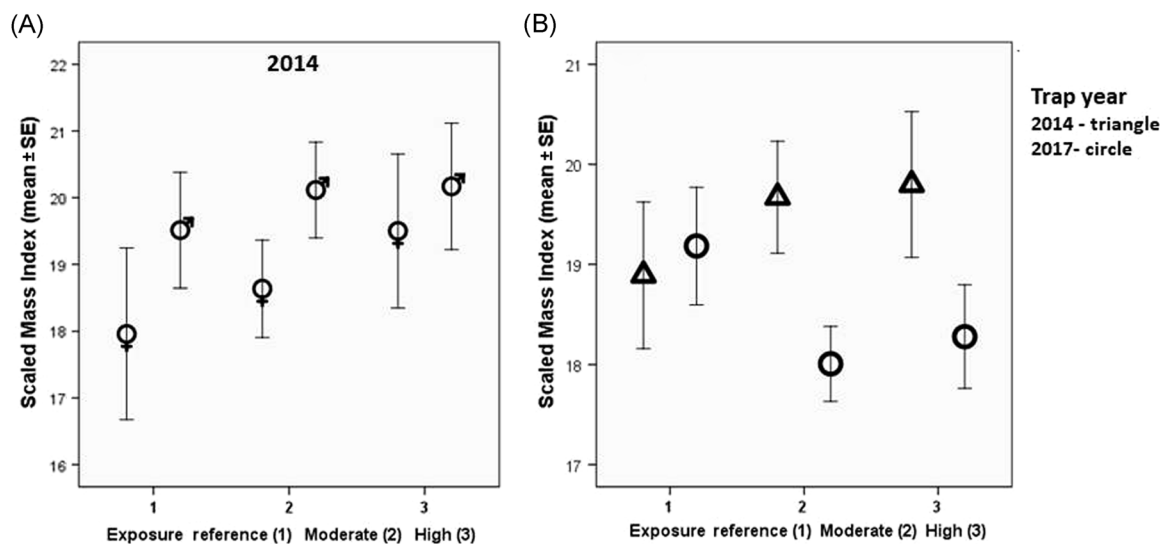
**FIGURE 1:** Induction of ethoxyresorufin-O-deethylase (EROD) enzyme from reference and impacted sites 2014 ($n = 40$) and 2017 ($n = 40$). In 2014, reference $n = 10$, moderate $n = 20$, and high $n = 10$.

correlated with Cr ($p = 0.027$); but when analyzed separately, only the males retained a positive and significant correlation. These differences disappeared by 2017.

DISCUSSION

Resident small mammals serve as sensitive and relevant bioindicators of environmental health (Shore 1995; Rodríguez-Estival and Smits 2016; Rodríguez-Estival et al. 2016) or, in this case, recovery of a contaminated ecosystem. In 2014, although there was no gross evidence of adverse effects on the voles, they were diverting energy toward actively detoxifying remnant contamination in the highly affected zones compared with those from the local reference sites. However, hepatic detoxification essentially returned to background/reference levels by 2017, providing evidence that toxicity from the bitumen-related compounds decreased over time through biodegradation, photolysis, and other natural processes.

The capture rate of the voles, used as a proxy for population density (Wobeser 1994), indicated that the vole numbers

**FIGURE 2:** Body condition (scaled mass index) of red-backed voles. In 2014, reference $n = 10$, moderate $n = 20$, and high $n = 9$. In 2017, reference $n = 10$, moderate $n = 20$, and high $n = 10$.

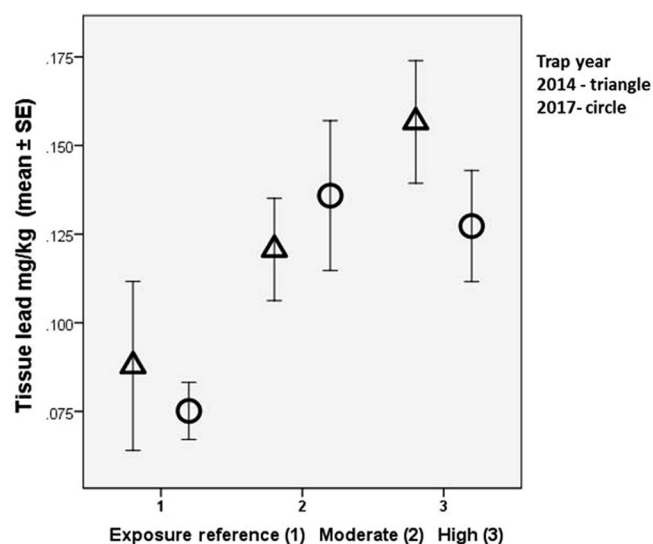


FIGURE 3: Tissue lead levels in red-backed voles in 2014 ($n = 37$) and 2017 ($n = 40$). In 2014, reference $n = 8$, moderate $n = 19$, and high $n = 9$. In 2017, reference $n = 10$, moderate $n = 20$, and high $n = 10$.

decreased to less than half in 2017 compared to 3 yr earlier. Southern red-backed vole populations are known to fluctuate and be cyclic (Elias et al. 2006; Fauteux et al. 2015). Fauteux et al. (2015) found population fluctuations of up to 41-fold. Based on EROD activity in the present study, polycyclic aromatic hydrocarbons and other xenobiotics associated with the spilled bitumen encountered by the voles through their food had returned to background levels (Figure 1). Considering this finding, together with capture rates at reference and impacted sites being similarly reduced, it is unlikely that the lower capture rate was a direct result of bitumen exposure. The lower capture rates could be attributable to several causes, none of which could be proven or

refuted: a natural fluctuation in the local population by the trapping and removal of voles in the course of these investigations, by higher predation rates, by higher overwinter die-offs, or because of less nourishing vegetation, none of which could be proven or refuted by the current study.

Given that Pb levels were higher in voles trapped on the affected sites and that it is 1 of the metals identified as increased in abiotic environmental samples downstream from the oil sands mining developments (Kelly et al. 2010), the higher tissue Pb levels likely are related to the release of bitumen, process-affected water, and steam. Lead levels were not discussed by Berger et al. (2016) because Pb concentrations in moss, herbaceous plants, and shrubs at impact sites did not exceed the concentrations detected in reference sites, by virtue of Pb levels being below detection levels at all sites. In addition, Pb in the soils of impacted sites did not exceed the concentrations observed in the reference sites or the concentrations outlined in the Alberta Tier 1 Soil and Ground Water Quality Remediation Guidelines (Alberta Environment and Sustainable Resource Development 2010). Therefore, soil Pb levels did not meet the criteria to be carried forward and assessed in the ERA (Intrinsik, Calgary, AL, Canada, unpublished data).

Voles with higher body burdens of Pb were in poorer body condition in 2014, a relationship that was evident from both the reference and affected areas but with a stronger correlation from the impact sites. Although Pb levels remained significantly higher in voles captured at affected sites compared to the reference site, the negative correlation with body condition was no longer evident in 2017. The source of that Pb is difficult to determine, but because it is a naturally occurring element, it may be concentrated, and/or exposed or released during various extraction processes. Although Pb is widely accepted to be a toxic metal, there is evidence that at low to moderate

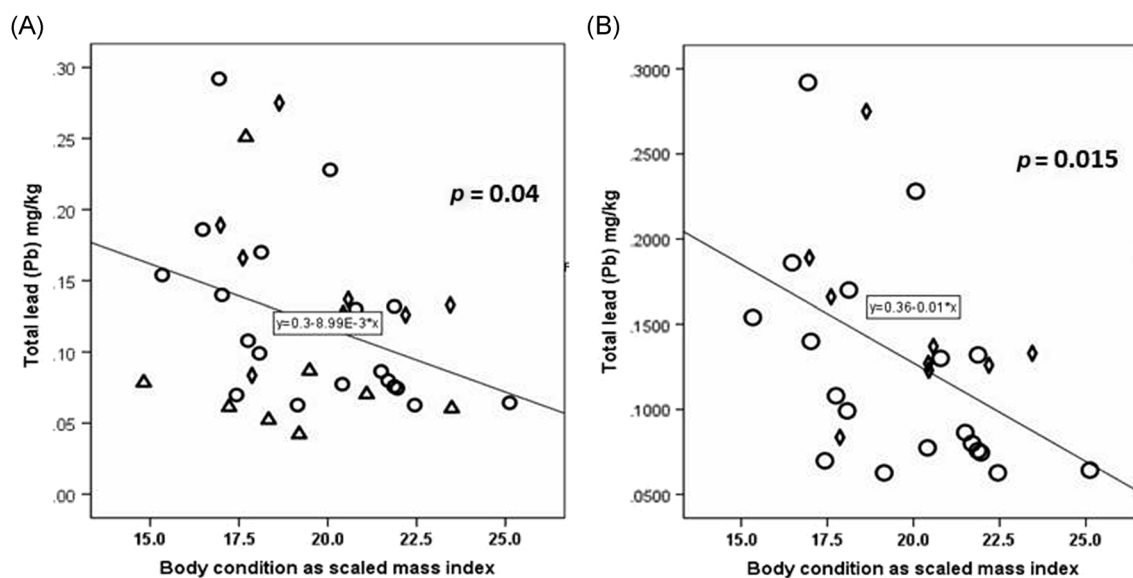


FIGURE 4: Body condition correlated with tissue lead in 2014. (A) Reference (triangle), moderate (circle), and high (diamond); $r^2 = -0.342$. (B) Oiled sites only; $r^2 = -0.457$.

levels Pb can have hormetic or measurably beneficial effects on the immune system particularly (Valentino et al. 2007; Mishra 2009; Nain and Smits 2011). Current research has also reported that wild sparrows from remote mining areas have evolved genetic adaptations to decrease their susceptibility to heavy metal toxicity (Andrew et al. 2019). The longer-term biological importance of the Pb levels seen in these animals is not known. This, as well as tracking Pb levels in other biota would be to determine if affected areas return to reference levels.

The higher Cr in voles from the most heavily affected sites in 2014 was a finding mirroring that of Berger et al. (2016); this mineral reverted to background levels similar to that of the reference site by 2017. The positive association seen with higher Cr and higher body condition may have been affected by other unknown, unmeasured factors. Chromium is an essential trace element for animals (Pechova and Pevlata 2007), so it is possible that the Cr at the spill sites was the essential cofactor, in its nontoxic trivalent form, which could be within the beneficial dose for voles. Speciation of the metals was beyond the scope of the present study.

Numerous factors affect and indicate ecosystem recovery. The present monitoring study followed 1 relevant, native biological indicator of ecosystem health and resilience, the southern red-backed vole. This species was identified through an earlier ERA (Intrinsik, Calgary, AL, Canada, unpublished data) as an appropriate bioindicator to use after a modified cleanup strategy was tested. Southern red-backed voles were selected as bioindicators because they are indigenous to the area and are likely to be exposed to COPC because of their home range size and herbivorous diet and because of the deposition of bitumen from the release (Intrinsik, Calgary, AL, Canada, unpublished data). As well, our work confirms that EROD remains a relevant and sensitive biomarker of contaminant exposure in sentinel small mammals, which also can point to the site recovery process. Wildlife can express a great range of “normal” for morphometrics because of a myriad of stochastic influences, so the SMI we used to reflect body condition allowed us to detect important relationships with exposure site and body levels of metals, which changed for the better over time.

Supplemental Data—The Supplemental Data are available on the Wiley Online Library at DOI: 10.1002/etc.4408.

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Data Accessibility—Data pertaining to the present study are available on request from J. Smits (judit.smits@ucalgary.ca).

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